BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: 4193

Retyped faithfully from the original board paper — same questions, no changes.

Q. Paper : II (Objective Type)

سوالیہ پرچہ: II (معروضی طرز)

QQ. 1. 1222 — Mathematics — (Inter Part - II) — (Session 2018-20 to 2020-22) — Paper (II) — (Objective)

Q. Time Allowed: 30 minutes — Maximum Marks: 20 — PAPER CODE 4193

Q1. $\int 2^x \, dx =$

(A) (B)

(C) (D)

Q2. The range of the function $f(x) = \sqrt{(x^2-4)}$ is,

(A) (B)

(C) (D)

Q3. $\lim_{x \to 0} (e^x-1)/x =$

(A) (B)

(C) (D)

Q4. $d/dx(\sqrt{(\tan x)}) =$

(A) (B)

(C) (D)

Q5. If $y = \sin\sqrt{x}$, then $dy/dx =$

(A) (B)

(C) (D)

Q6. d/dx(1/(1/ℓnx)) =

(A) (B)

(C) (D)

Q7. d/dx(1/cosec x) =

(A) (B)

(C) (D)

Q8. $\int x^{-1}\, dx$ =

(A) (B)

(C) (D)

Q9. The direction cosines of a vector $3\underline{i} - \underline{j} + 2\underline{k}$ are

(A) (B)

(C) (D)

Q10. The solution of the differential equation $dy/dx = (y^2+1)/e^{-x}$ is,

(A) (B)

(C) (D)

Q11. $\int(\cot x)/(\ell n \sin x)\, dx$ =

(A) (B)

(C) (D)

Q12. If a line ℓ is perpendicular to x-axis, then its inclination is,

(A) (B)

(C) (D)

Q13. The equation of the straight line whose slope is 2 and y-intercept 5 is,

(A) (B)

(C) (D)

Q14. The distance of a point P(6, −1) from the line 6x − 4y + 9 = 0 is

(A) (B)

(C) (D)

Q15. The slope of line through the points (−2, 4), (5, 11) is

(A) (B)

(C) (D)

Q16. Point (3, 2) is not the solution of inequality

(A) (B)

(C) (D)

Q17. The focus of the parabola $x^2 = 8y$ is

(A) (B)

(C) (D)

Q18. The eccentricity of the hyperbola is

(A) (B)

(C) (D)

Q19. The Centre of the circle $(x-1)^2 + (y+3)^2 = 3$ is

(A) (B)

(C) (D)

Q20. Which one of the following is not a unit vector,

(A) (B)

(C) (D)

Handwritten "SGD-22" after the paper code is not transcribed. The stem of Q14 prints 'i4)' (typo) verbatim.

Paper : II (Essay Type)

پرچہ: II (انشائیہ طرز)

QSection I. 1222 — Mathematics — (Inter Part - II) — (Session 2018-20 to 2020-22) — Paper (II) — (Subjective)

Q. Time Allowed: 2.30 hours — Maximum Marks: 80

Q. Section ---------- II — (10 × 3 = 30)

Q. Note: Attempt any three questions.